

Mount Kenya



University

UNIVERSITY EXAMINATION 2018/2019

**SCHOOL OF ENGINEERING, ENERGY AND THE BUILT ENVIRONMENT
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING**

**DEE/ DME
REGULAR**

UNIT CODE: DEG1103

UNIT TITLE: BASIC ELECTRONICS I

TUESDAY 20TH NOVEMBER, 2018 11.00AM MAIN EXAM TIME: 2 HOURS

INSTRUCTIONS:

This paper consist of two sections, question one in section A is compulsory attempt any other two questions in section B

SECTION A

1. a) Define;
 - i) Doping
 - ii) Electron emission
 - iii) Work function(3 Marks)
- b) With aid of a sketch, describe photo emission (5 Marks)
- c) Explain the meaning of each of the following
 - i) extrinsic semiconductor
 - ii) Intrinsic semiconductor(4 Marks)
- d) i. With aid of a diagram explain the operation full wave centre tap rectifier amplifier (6 Marks)
 - ii. Sketch the input and output waveforms of voltage for the rectifier in c (i) above. (3 Marks)

- e) Draw the symbols for
- i) semiconductor diode
 - ii) LED
 - iii) Directly heated cathode (3 Marks)
- f) Explain the difference between negative and positive coefficient of resistance. (2 Marks)
- g) Explain the effect of temperature on semiconductors. (4 Marks)

SECTION B

2. a) With aid of a diagram, explain the formation of n type semiconductor. (7 Marks)
- b) Describe the valence band, conduction band and forbidden energy gap with the help of energy level diagram. (9 Marks)
- c) Describe how you would determine the terminal of a LED using the multimeter. (4 Marks)
3. a) explain three properties of semiconductors (6 Marks)
- b) With aid of a diagram explain the VI characteristics of a pn diode (7 Marks)
- c) A half-wave rectifier is used to supply 50V d.c. to a resistive load of $800\ \Omega$. The diode has a resistance of $25\ \Omega$. Calculate a.c. voltage required. (7 Marks)
4. a) Define;
- i. reverse biasing
 - ii. Forward biasing (2 Marks)
- b) Sketch the diagram of a cathode ray tube and label all the parts (5 Marks)
- c) Highlight three properties of thermionic emitter (3 Marks)

d) A full-wave rectifier uses two diodes, the internal resistance of each diode may be assumed constant at $20\ \Omega$. The transformer r.m.s. secondary voltage from centre tap to each end of secondary is 50 V and load resistance is $980\ \Omega$. determine

- i) The mean load current
- ii) The r.m.s. value of load current (7 Marks)

e) Highlight any three applications of diodes (3 Marks)

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